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<Design of a short-range remote methane
detector>

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Final Year Project Interim Report

Submitted in Partial Fulfilment of the
Requirements for
The Degree of Bachelor of Engineering

Department of Mechanical and Automation
Engineering

The Chinese University of Hong Kong

December 2025

Abstract

1. This report presents the design of a remote methane detector using the principles and knowledge of optical spectroscopy and gas diagnostics as detailed in the textbook “Spectroscopy and Optical Diagnostics for Gases” by Ronald K. Hanson, R. Mitchell Spearrin, and Christopher S. Goldenstein.
2. This project aims to design the remote methane detector for bringing a convenient solution to detect the hidden gas in the industrial environment and monitoring the industrial safety.
3. The testing and lab experiment will be using the gas methane (CH_4), which is a greenhouse gas and key component some of the industrial environment, for example, construction site for building, gas station.

Acknowledgement

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Problem description

1. Methane, work as an invisible, odourless and highly flammable gas, that works as a key component of the natural gas, as the natural gas works and exist in various site, for example, construction site, gas station, etc. Which will increase the risk of the people who works and living nearby, such, providing a solution for detecting the methane leaking is a serious and important thing for protecting and helping the society.
2. As compared to the traditional solution for detecting methane leaking, soap bubbling test, laser leak detector [1], the remote detector use in this project is more convenient and the function will be more perfect, as soap bubbling only can test with locating the connections and pipes, and laser leak detector only works for methane [1], shows that they will have many limitations that limit their usage and functions, but the remote methane detector for this project works if the air located at the suspicious leaking site can insert to the pipe to detect the leakage. also, while we change the IR laser and the wedge window to a suitable wavelength, we also can detect some of the other toxic gases if they are also leaking in there.

Result of literature review and Background Knowledge

1. The system mainly utilizing with the infrared laser source (IR), gas cell system, and also with the receiver, by the laser beam passes through the gas cell, the gas in the cell will absorb the IR laser, by the vibrational transitions in heteronuclear molecules, and while the vibration causing a change in the electric dipole moment [2], which will make the gas to absorb the laser, and while in this moment, we can use the receiver to represent the data with the intensity or power vs the time or the voltage of the receiver, to show whether the gas exist in the gas cell or not.

The design of the system cell, the design of the system cell includes, the left and right side will have 2 mounting plates to mount the window and the O-ring to have a double O-ring face seal situation for keeping the system in vacuum and isolate the gas with air, to prevent the gas in the air affect the data collected In the cell.

While at the top of the system there are two NPT type thread hole, which is for the male connector [3], for inserting and releases the gas in and out of the cell.

At the bottom there are two M4 size thread, which is use for holding the entire cell, to make it more stable while using the cell to do the experiment.

2. For the material use, first the material of the cell use will be Aluminium, there are several to choose to use Aluminium as the material of the gas cell.

First, aluminium can be a high-cost performance ratio of the solution to the gas cell, since the aluminium can be customized the design using 3d drawing

software, for example fusion 360, SolidWorks, etc. And compare to other material to fabricate, stainless steel will be too expensive, and the PLA, ABS 3d printing material, the dimension will have a much more error compare to Al13 and cannot ensure the air permeability of the material and the material status while in the vacuum environment, so, to choose to use aluminium is an alternative choice compare to stainless steel and 3d printing material, with more suitable to this project use.

Second, aluminium has a good thermal conductivity, which is amount $237 \text{ Wm}^{-1}\text{K}^{-1}$, and low density compare to steel and cast iron [4], which means that the aluminium will have a good heat transfer rate, in some of the situation, for example, hot construction site, it can transfer the heat out to the surroundings quickly, and the temperature may increase the energy and intensity power of the gas, by formula $KE = 3/2RT$ shows, the temperature increase, the energy of gas will also increase, and it will affect the amount of energy that the molecules can absorb, since laser also is a type of energy sources, which will makes the molecules absorb less laser, and affect the accuracy of intensity power of laser that collect and represent using the receiver, such that using aluminium is a good choice for gas cell since have a great thermal conductivity

3. For the glass use in the gas cell, there will be using 2 high accuracy wedge windows, the high accuracy is needed for stabilize the laser input and output, to avoid any holes on the glass and the dimension is correct with less errors, also since we are choose to use the wedge glass, which the surface of the wedge glass are at a slight angle, to prevent the reflection of the laser back to the surroundings, as compare to the normal glass, the flat surface of the normal glass will reflect some of the laser back to the source site [5], such compare to wedge glass, using normal glass in this project may affect the accuracy of the data, such wedge glass is more preferred.
4. For the O-ring dimensions and the material use for the gas cell, due to squeezing of the force with the O-ring, the O-ring ID should 1-5% slight smaller than the groove and the hardware (wedge glass) diameter [6], and the OD of the O-ring is better to $\geq$ the groove OD, such, since the glass choosing is around 12.5mm, then by the formula, the O-ring ID is at most 12.125mm and the OD is more depends on the thickness of the O-ring. And by choosing the correct O-ring, from Taobao, which is 17OD x12.2ID and 2.4mm thickness, which can help to seal the glass and the tube to make it vacuum.

For the way of sealing for the gas cell, axial face sealing is using due to the mounting plate and the gas cell, so that two O-ring can push together with the

compression force provide by the screw and the mounting plate, will makes the connection parts of the wedge window and the cell make vacuum [7], to isolate the air to goes in the pipe.

For the height of the groove, since we are using axial face sealing, such, 20% or 30% squeeze is recommended [7], and by the formula of recommend groove height, $\text{squeeze\%} = (\text{Initial_Height} - \text{Squeeze_Height}) / \text{Initial_Height}$ [8], such that, after choose the 12.2mm O-ring, the height is 2.4mm and the squeeze percentage is 25%, then the groove height should be around $25\% = (2.4 - \text{SH}) / 2.4$, and SH should be 1.8mm, which the height is decided for the O-ring and apply in the cad drawing.

For the material use of O-ring, according to [9], since we are assuming the gas tube will in vacuum situation, such Viton and Aflas type O-ring we are using.

Background knowledge:

- 1.

Approach to the problem

1. Select the IR laser source, as methane absorb specific wavelength of laser sources, so selecting the specific IR laser will be very important, as the research shows, the absorption line of methane wavelength range is from 1.65, 2.3, 3.33.4 and 7.7 μm [10], so while we doing the experiment, we need to use the IR laser with wavelength range from above.
2. Select the receiver for the IR laser, since the IR laser for the methane works with specific wavelength, so while we select the specific laser, choose appropriate receiver should also be one of the approaches to detect the intensity of the laser inside the gas cell.
3. Design and assemble the testing gas cell, the gas cell is asked to not exceed the length of 15 cm, so the design is based on ≤ 15 cm after assembling for the whole gas cell design, the whole gas cell includes 3 main parts. 1. The main gas cell with connected stand and male connector. 2. The 2 mounting plates with groove and 4 O-rings and wedge window. 3. The IR laser and Receiver.
4. Testing using the application, with the receiver receive the intensity of the laser inside the gas cell and present them using the data or plot graph.
5. Validating the results using formula and find if there is any error or optimization can do with the system.
6. Optimize the system with checking the vacuum of the cell, and the fittings if connected perfectly, and see if any upgrade of the cell can be made.

Results and discussions

Result:

1. Design of the methane gas cell.
2. The result observed after experiment and testing.
3. Combined results using matlab.

Discussions

- 1.

Budget Control

1. Order CNC fabrication of the gas cell and mounting plate.
2. 4 x O-ring with specific size [11].
3. 2 x wedge window with specific coating film [12].
4. 1 x Swagelok PTFE Tape Thread Sealant for sealing for male connector [13].
5. 1 set of M5 x 20mm screw for connecting mounting plate and gas cell [14].
6. 1 set of M4 x 10 mm screw for connecting the stand and the gas cell [15].
7. 1 x Swagelok male connector Straight Tube Fitting [3].
8. 1 x Stainless Steel Quarter Turn Instrument Plug Valve [16].
9. 1 x Swagelok Stainless Steel Swagelok Tube Fitting [17].

Conclusion

References

- [1] Climate and Clean Air Coalition (CCAC) Oil and Gas Methane Partnership, "Methane Emissions Detection and Measurement Techniques, Equipment and Costs," Apr. 2016. [Online]. Available: <https://www.epa.gov/sites/default/files/2016-04/documents/mon7ccacemissurvey.pdf>
- [2] R. K. Hanson, R. M. Spearrin, and C. S. Goldenstein, *Spectroscopy and Optical Diagnostics for Gases*, Cham, Switzerland: Springer International Publishing, 2016, doi: 10.1007/978-3-319-23252-2.
- [3] Swagelok, "SS-400-1-4 Straight Tube Fitting," Swagelok Products, 2024. [Online]. Available: <https://products.swagelok.com/en/c/straights/p/SS-4001-4>. [Accessed: 26-Nov-2025].
- [4] A. Zhang and Y. Li, "Thermal Conductivity of Aluminum Alloys—A Review," *Materials*, vol. 16, no. 2972, Apr. 2023, doi: 10.3390/ma16082972.
- [5] Manx Precision Optics Ltd., "Etalons Explained," Technical Note 202006, Manx Precision Optics Ltd., 2020. [Online]. Available: <https://mpo.im/wp-content/uploads/2020/06/TN-2020-06-Etalons-Explained.pdf>
- [6] V. WayKen, "O-Ring Groove (Gland) Design: A Detailed Guideline," WayKen Rapid Manufacturing, May 19, 2023. [Online]. Available: <https://waykenrm.com/blogs/o-ring-groove-design-guide/>. [Accessed: 29Nov-2025].
- [7] Marco Rubber and Plastics, "Understanding O-Ring Squeeze, Stretch & Compression," Marco Rubber, 2025. [Online]. Available: <https://www.marcorubber.com/o-ring-groove-design-considerations.htm>. [Accessed: 30-Nov-2025].
- [8] "How do you calculate O-ring compression force?," Engineering Product Design, Dec. 19, 2022. [Online]. Available:

<https://engineeringproductdesign.com/knowledge-base/o-ring-compression-force/>. [Accessed: 30-Nov-2025].

- [9] "O-Ring Material Chart - Rubber Selection Guide Quick Reference," Marco Rubber & Plastics. [Online]. Available: <https://www.marcorubber.com/o-ring-material-quick-reference.htm>. [Accessed: 30-Nov-2025].
- [10] S. V. Yakovlev, O. A. Romanovskii, S. A. Sadovnikov, D. A. Tuzhilkin, A. A. Nevzorov, O. V. Kharchenko, and N. S. Kravtsova, "Mobile midinfrared differential absorption lidar for methane monitoring in the atmosphere: Calibration and first in situ tests," **Results in Optics**, vol. 8, p. 100233, Apr. 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2666950122000232>
- [11] Tmall, "GW14-K9 高精度楔形窗口 未镀膜 光学元件 实验器材科研 激光谐振腔光学干扰保护楔窗口 K9 光学玻璃," [Online]. Available: <https://detail.tmall.com/item.htm?id=789620038856>. [Accessed: 29-Nov2025].
- [12] Tmall, "12.9 级高强度杯头内六角螺丝钉螺母组合套装大全螺栓平 弹垫 M2-M20," [Online]. Available: <https://detail.tmall.com/item.htm?id=631928248397>. [Accessed: 29-Nov2025].
- [13] Tmall, "12.9 级高强度镀镍内六角螺丝杯头螺栓螺丝钉螺杆 M2M3M4M5M6M10M12," [Online]. Available: <https://detail.tmall.com/item.htm?id=20471255293>. [Accessed: 29-Nov2025].
- [14] Taobao, "美国 Swagelok 世伟洛克 MS-STR-8 生料带 MS-STR-4 封 带螺纹密封剂 PTFE," [Online]. Available: <https://item.taobao.com/item.htm?id=816501930703>. [Accessed: 29-Nov2025].

- [15] Tmall, "氟胶 O 型圈线径 1.5/2/2.4/3.1/3.5/4/4.5/5 骨架油封耐高温密封圈," [Online]. Available:
<https://detail.tmall.com/item.htm?id=732800793944>. [Accessed:
30-Nov2025].
- [16] Swagelok, "SS-4P4T-BK 2-Way Straight Pattern Valve," Swagelok Products, 2024. [Online]. Available:
<https://products.swagelok.com.cn/en/c/2-way-straight-pattern/p/SS-4P4TBK>. [Accessed: 26-Nov-2025].
- [17] Swagelok, "SS-400-6SC11 Straight Tube Fitting," Swagelok Products, 2024. [Online]. Available:
<https://products.swagelok.com.cn/en/c/straights/p/SS-400-6SC11>.
[Accessed:
26-Nov-2025].